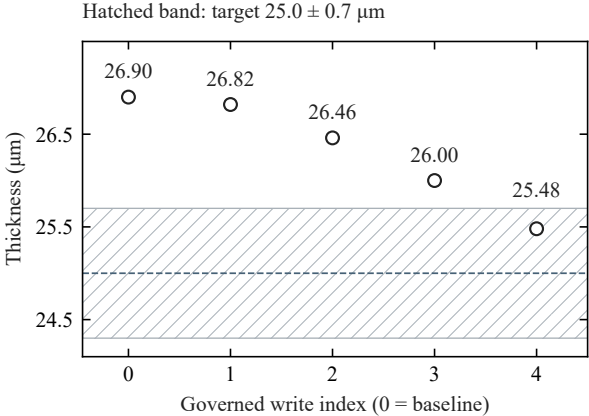


(a) Archived thickness observations



(b) Governed-write ledger

Write	Parameter / setpoint change	Transport
1	Casting speed 32.0 → 33.1 m/min	Modbus TCP
2	MDO fast roll 118.0 → 120.8 m/min	Modbus TCP
3	TDO outlet width 3000.0 → 3050.0 mm	OPC UA
4	Casting speed 33.1 → 33.7 m/min	Modbus TCP

4 records; logged keep; parent COMPLETED.

Frozen B; simulated plant; scripted policy. Markers are returned-bucket means, not a continuous trajectory.